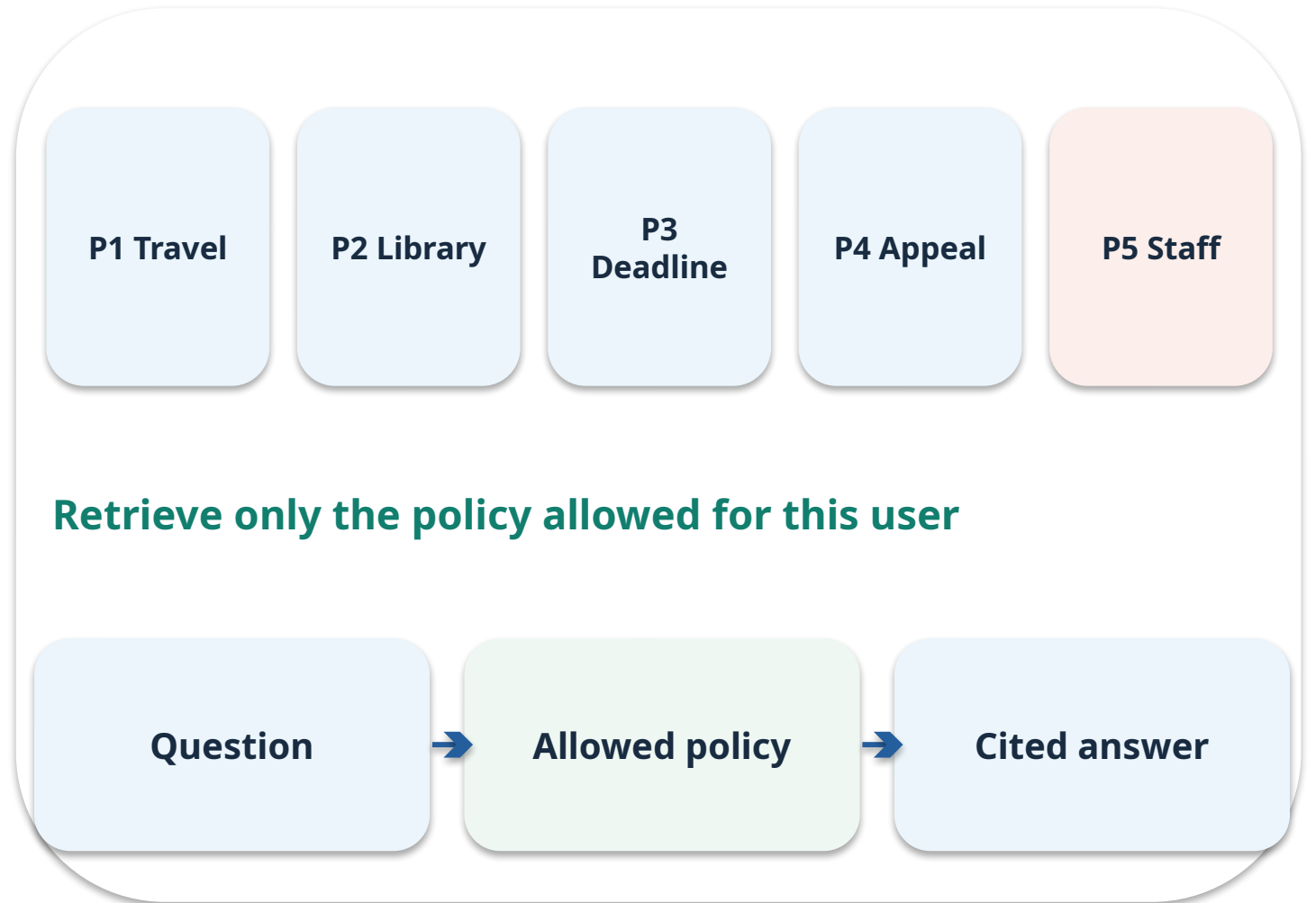


Week 07 · Building and checking RAG

OPEN / WEEK 07

The cited deadline answer

The assistant cites P3 for “two extensions,” but P3 permits one. A citation is present. Is the answer supported?



First vote: what would you do?

QUESTION / WEEK 07

The assistant cites P3 for “two extensions,” but P3 permits one. A citation is present. Is the answer supported?

Choose an initial answer and name one piece of evidence you need.

VOTE FIRST • GIVE A REASON

A Accept the first plausible result

B Check the source or measurement

Which source or test could settle this?

The journey through today's case

OPEN / WEEK 07

We will predict, connect an analogy, trace a mechanism, work an example, then explain a decision.

1

RAG

2

Citation support

3

No-answer case

4

Test split

1. Predict · RAG

QUESTION / WEEK 07

Which step happens before drafting?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A Find the relevant allowed passage.

B A citation added afterward repairs a false answer.

Which source or test could settle this?

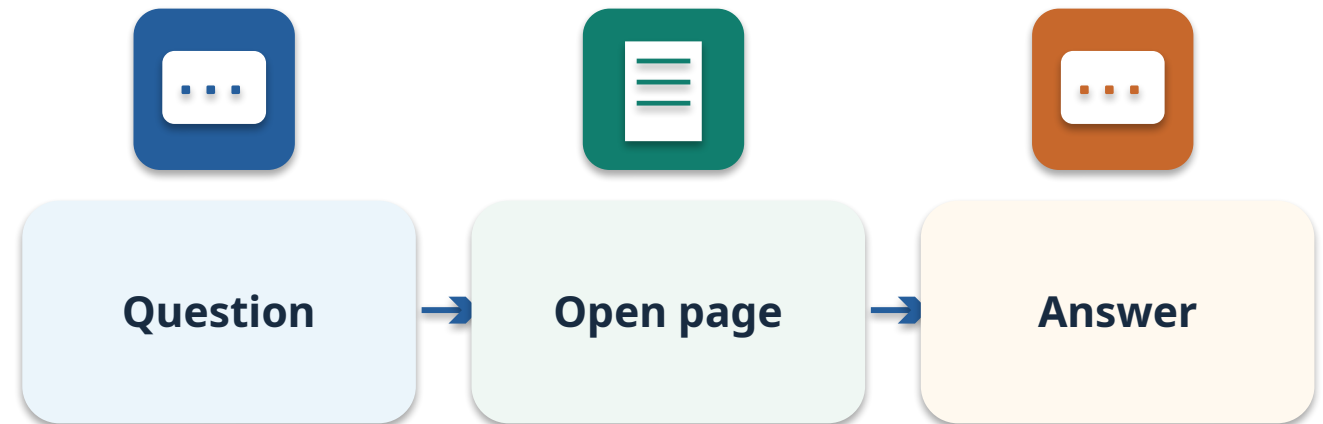
RAG · a familiar picture

ANALOGY / WEEK 07

An open-book examiner finds the page before writing the answer.

What matches our case, and where does the analogy stop?

An open-book examiner finds the page before writing the answer.



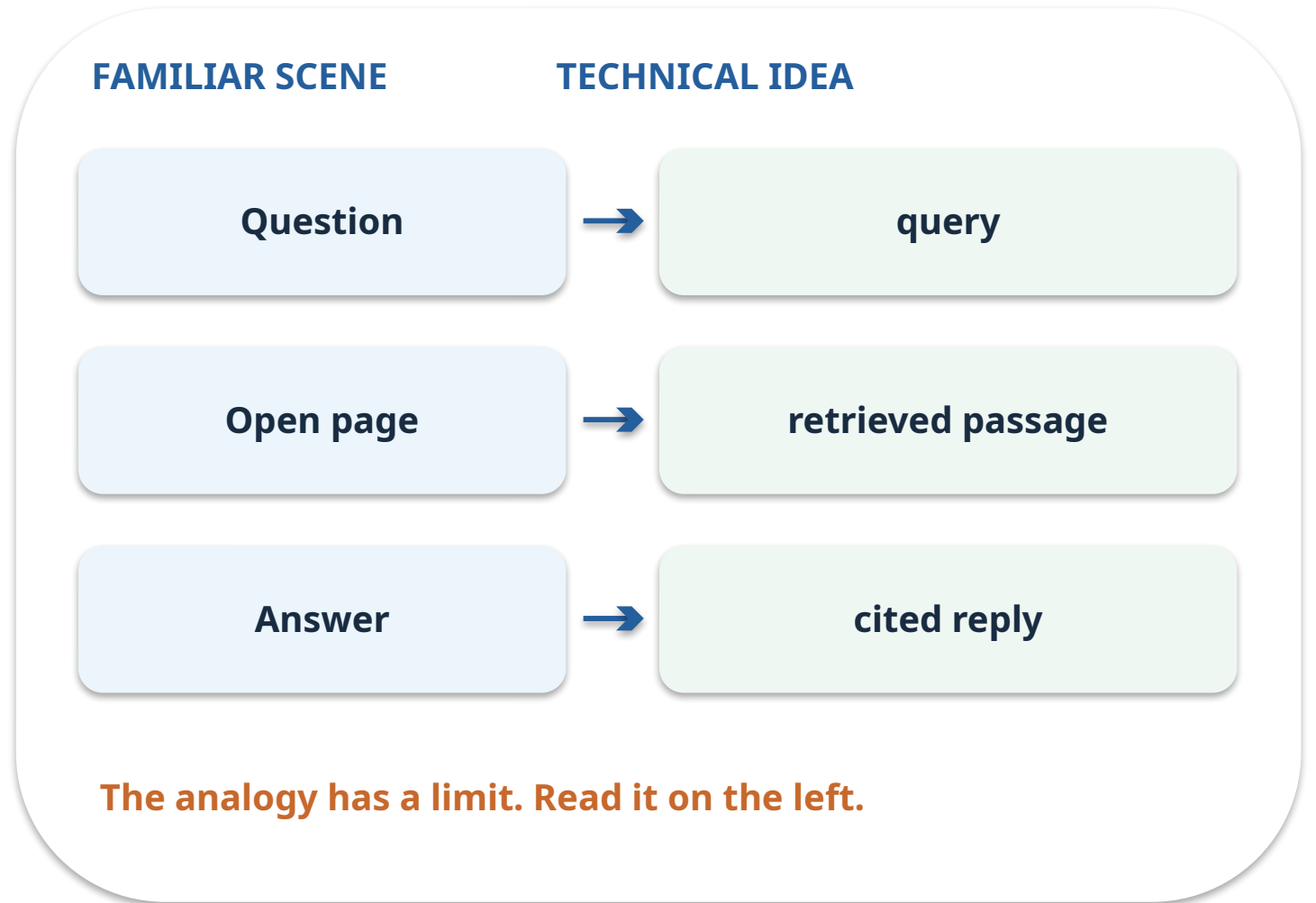
What matches the technical process?

RAG · map the parts

MAPPING / WEEK 07

Match each everyday part to the technical part.

Analogy limit: Retrieval can pick an irrelevant page; answer quality depends on what was found.



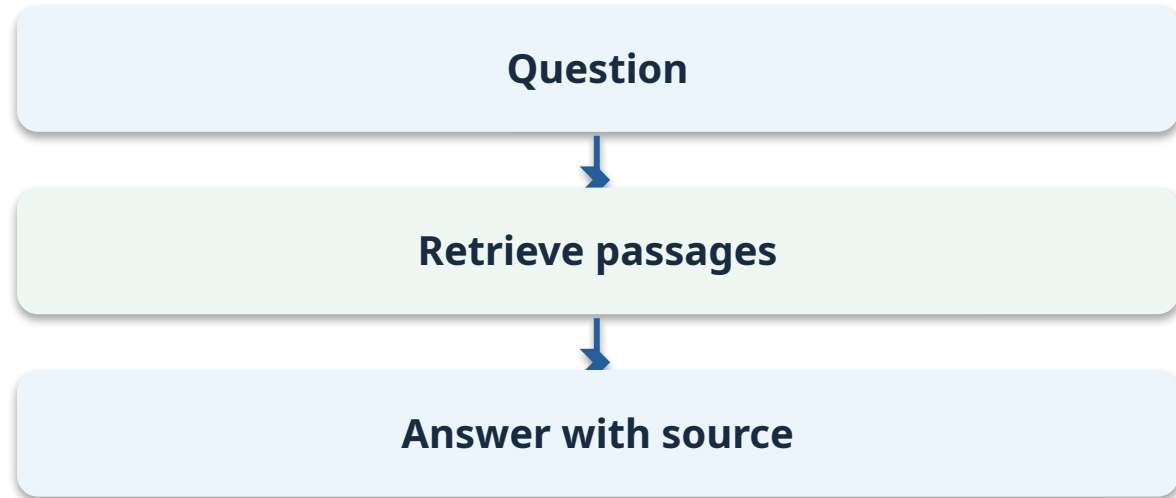
RAG · how it works

MECHANISM / WEEK 07

Retrieve relevant text for the current request, then draft an answer bounded by that text.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



RAG · work one case

WORKED / WEEK 07

P3 states one extension in seven days; answer “one [P3]” after retrieving P3.

Point to the evidence before giving your conclusion.

GIVEN

P3 states one extension in seven days; answer “one [P3]” after retrieving P3.

QUESTION

Which step happens before drafting?

SUPPORTED ANSWER

Find the relevant allowed passage.

RAG · answer and explain

REVEAL / WEEK 07

Find the relevant allowed passage.

Tempting mistake: A citation added afterward repairs a false answer.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

A citation added afterward repairs a false answer.

SUPPORTED EXPLANATION

Find the relevant allowed passage.

Retrieval can pick an irrelevant page; answer quality depends on what was found.

2. Predict · Citation support

QUESTION / WEEK 07

What must be checked beyond the presence of a citation?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A The passage must support the exact claim.

B Any source ID proves the sentence.

Which source or test could settle this?

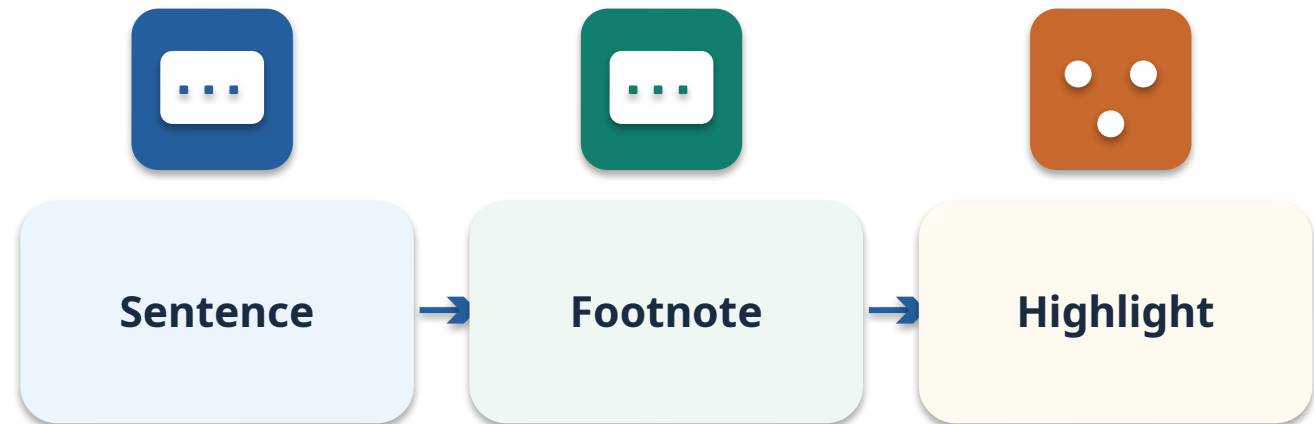
Citation support · a familiar picture

ANALOGY / WEEK 07

A footnote is useful only when its source actually says the claim.

What matches our case, and where does the analogy stop?

A footnote is useful only when its source actually says the claim.



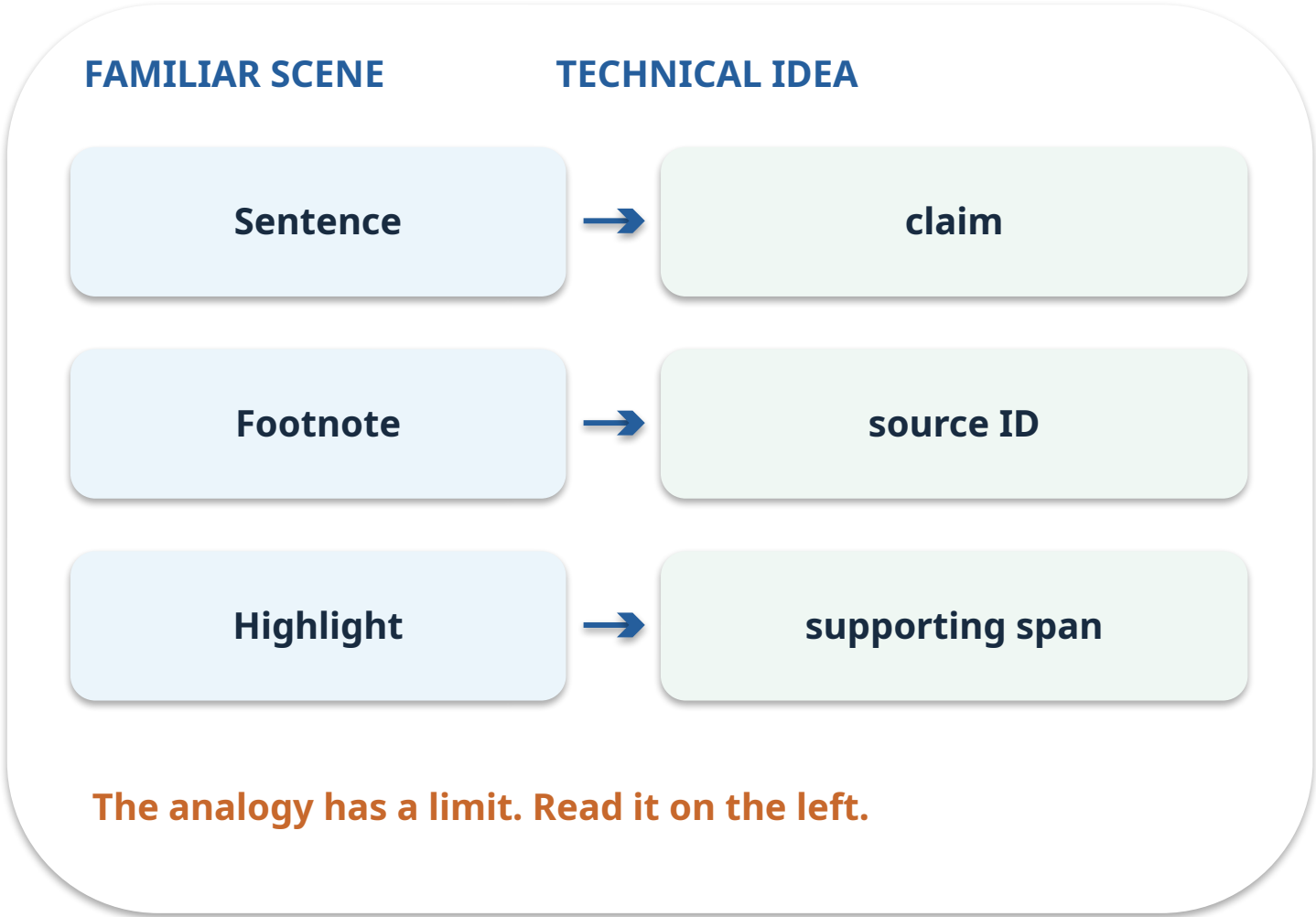
What matches the technical process?

Citation support · map the parts

MAPPING / WEEK 07

Match each everyday part to the technical part.

Analogy limit: Checking entailment can involve nuance; use a human for disputed cases.



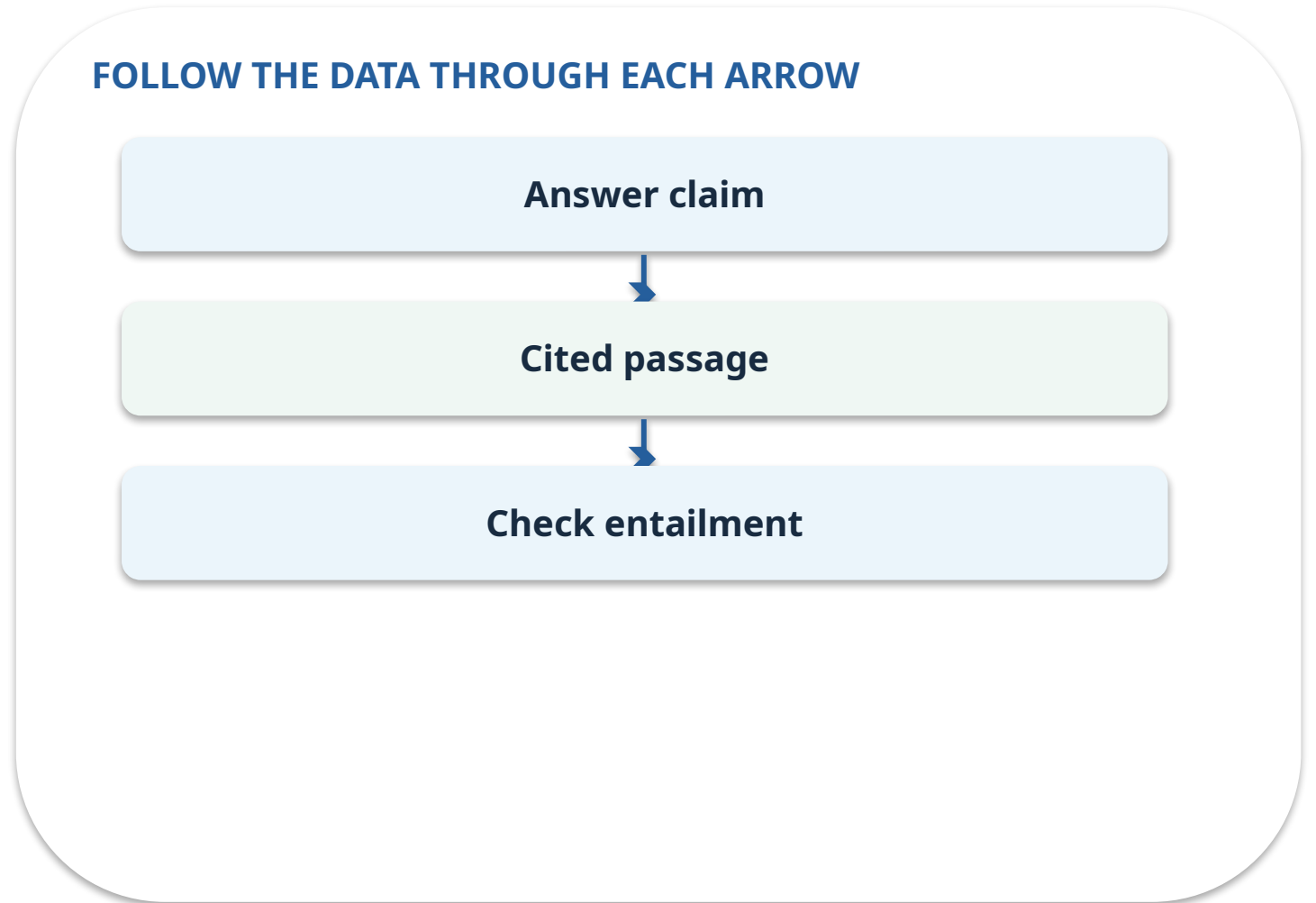
Citation support · how it works

MECHANISM / WEEK 07

Check every material claim against the cited passage.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Citation support · work one case

WORKED / WEEK 07

“Two extensions [P3]” is unsupported because P3 says one, even though the ID exists.

Point to the evidence before giving your conclusion.

GIVEN

“Two extensions [P3]” is unsupported because P3 says one, even though the ID exists.

QUESTION

What must be checked beyond the presence of a citation?

SUPPORTED ANSWER

The passage must support the exact claim.

Citation support · answer and explain

REVEAL / WEEK 07

The passage must support the exact claim.

Tempting mistake: Any source ID proves the sentence.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

Any source ID proves the sentence.

SUPPORTED EXPLANATION

The passage must support the exact claim.

Checking entailment can involve nuance; use a human for disputed cases.

3. Predict · No-answer case

QUESTION / WEEK 07

What should the system say?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A It lacks a relevant approved source in this corpus.

B The model should guess from common knowledge.

Which source or test could settle this?

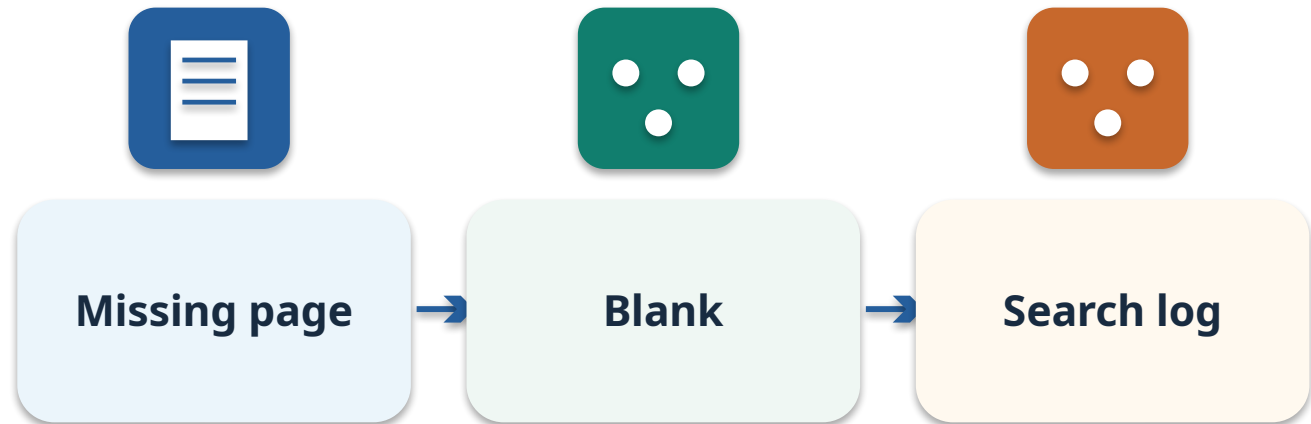
No-answer case · a familiar picture

ANALOGY / WEEK 07

A careful researcher leaves a blank when the archive lacks evidence.

What matches our case, and where does the analogy stop?

A careful researcher leaves a blank when the archive lacks evidence.



What matches the technical process?

No-answer case · map the parts

MAPPING / WEEK 07

Match each everyday part to the technical part.

Analogy limit: A retrieval miss may hide a real answer elsewhere; report the search limit.

FAMILIAR SCENE

Missing page

Blank

Search log

TECHNICAL IDEA

no relevant passage

abstain

reason

The analogy has a limit. Read it on the left.

No-answer case · how it works

MECHANISM / WEEK 07

If approved sources lack an answer, return a clear unsupported status.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW

No relevant source



Abstain



Explain missing evidence

No-answer case · work one case

WORKED / WEEK 07

Ask about parking permits;
none of five policies covers it, so
say evidence is unavailable.

Point to the evidence before
giving your conclusion.

GIVEN

**Ask about parking permits; none of five policies covers it, so
say evidence is unavailable.**

QUESTION

What should the system say?

SUPPORTED ANSWER

It lacks a relevant approved source in this corpus.

No-answer case · answer and explain

REVEAL / WEEK 07

It lacks a relevant approved source in this corpus.

Tempting mistake: The model should guess from common knowledge.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

The model should guess from common knowledge.

SUPPORTED EXPLANATION

It lacks a relevant approved source in this corpus.

A retrieval miss may hide a real answer elsewhere; report the search limit.

4. Predict · Test split

QUESTION / WEEK 07

Why not tune again after seeing the ten final answers?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A That would contaminate the independent check.

B Reusing final questions improves an unbiased estimate.

Which source or test could settle this?

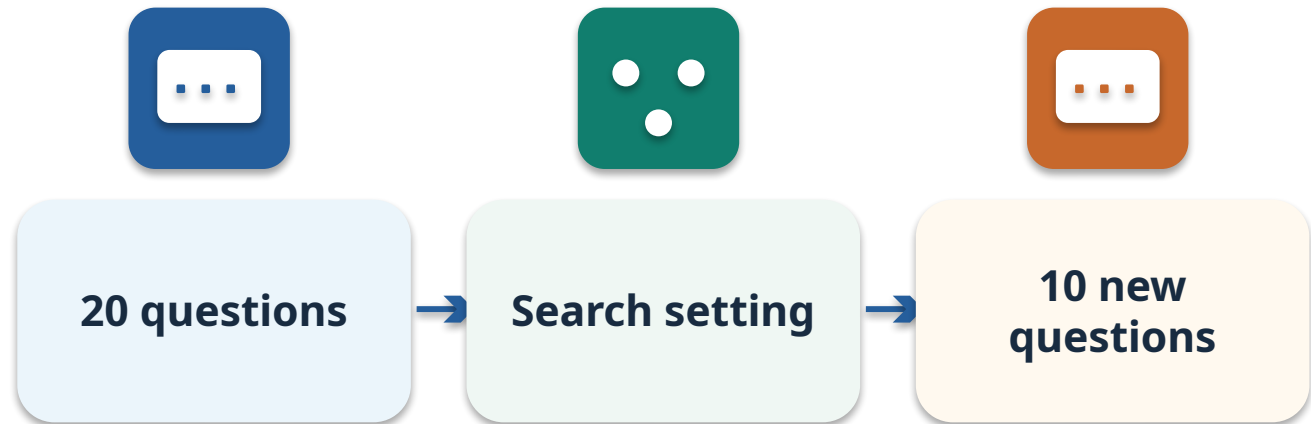
Test split · a familiar picture

ANALOGY / WEEK 07

A practice quiz guides choices;
an unopened exam checks the choice.

What matches our case, and
where does the analogy stop?

A practice quiz guides choices; an unopened exam checks the choice.



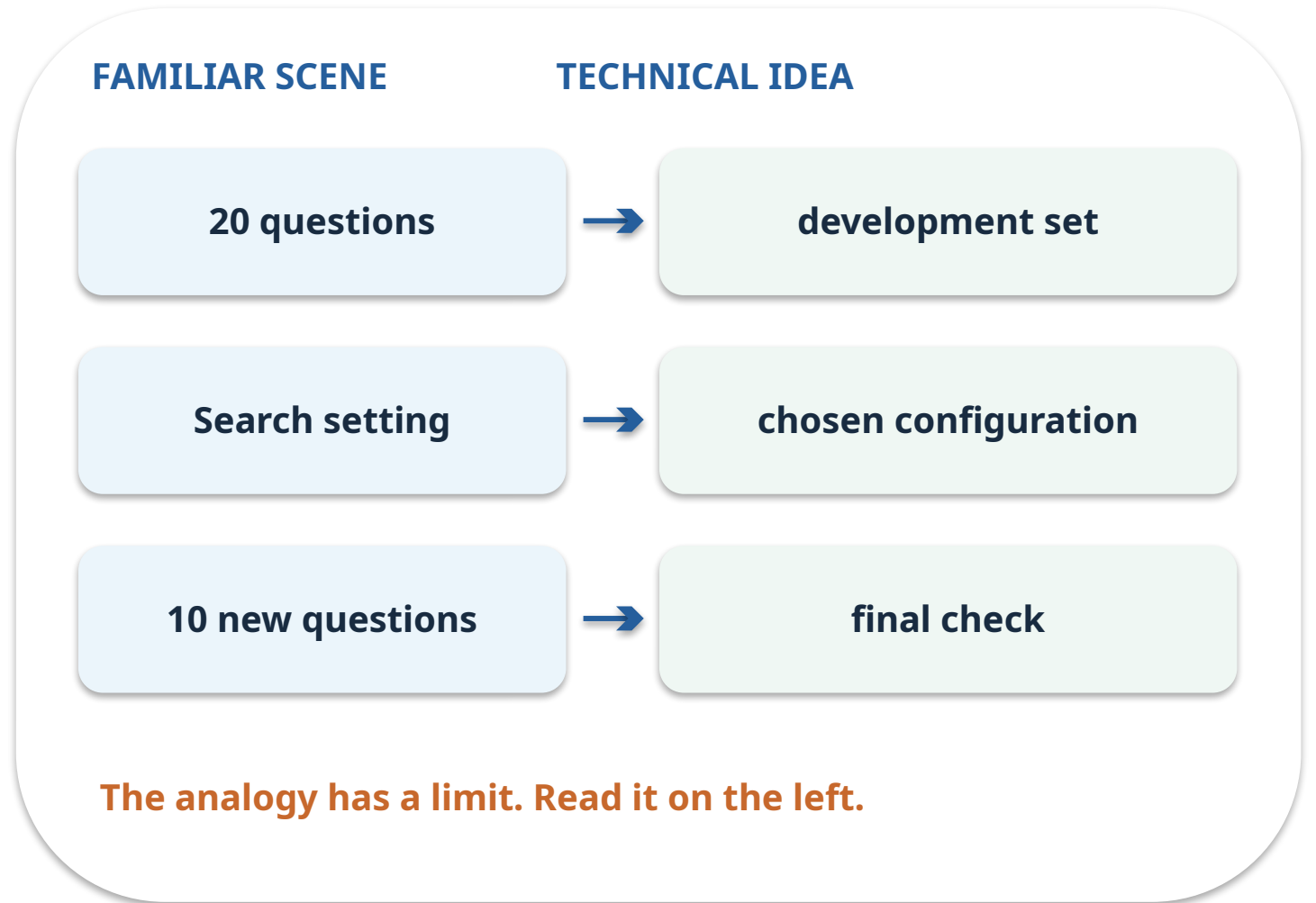
What matches the technical process?

Test split · map the parts

MAPPING / WEEK 07

Match each everyday part to the technical part.

Analogy limit: Ten final cases are still a small sample; report each failure and denominator.



Test split · how it works

MECHANISM / WEEK 07

Use 20 practice questions for improvements and keep ten separate questions for final checking.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Test split · work one case

WORKED / WEEK 07

Compare top-k 1 and 2 on 20 practice questions, freeze the choice, then open ten final cases.

Point to the evidence before giving your conclusion.

GIVEN

Compare top-k 1 and 2 on 20 practice questions, freeze the choice, then open ten final cases.

QUESTION

Why not tune again after seeing the ten final answers?

SUPPORTED ANSWER

That would contaminate the independent check.

Test split · answer and explain

REVEAL / WEEK 07

That would contaminate the independent check.

Tempting mistake: Reusing final questions improves an unbiased estimate.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

Reusing final questions improves an unbiased estimate.

SUPPORTED EXPLANATION

That would contaminate the independent check.

Ten final cases are still a small sample; report each failure and denominator.

Evidence trail: input → result

WORKED / WEEK 07

Q: How many extensions are allowed?

A: One deadline extension [P3].

GIVEN

P3: Students may request one deadline extension within seven days.

QUESTION

Q: How many extensions are allowed?

SUPPORTED ANSWER

A: One deadline extension [P3].

Notebook: run the prepared example

PRACTICE / WEEK 07

Run a small extractive retrieval and answer pipeline.

Read the original output before editing.

1

Search for a relevant policy

2

Produce an extractive classroom answer with a source ID

3

Record supported and unsupported questions

4

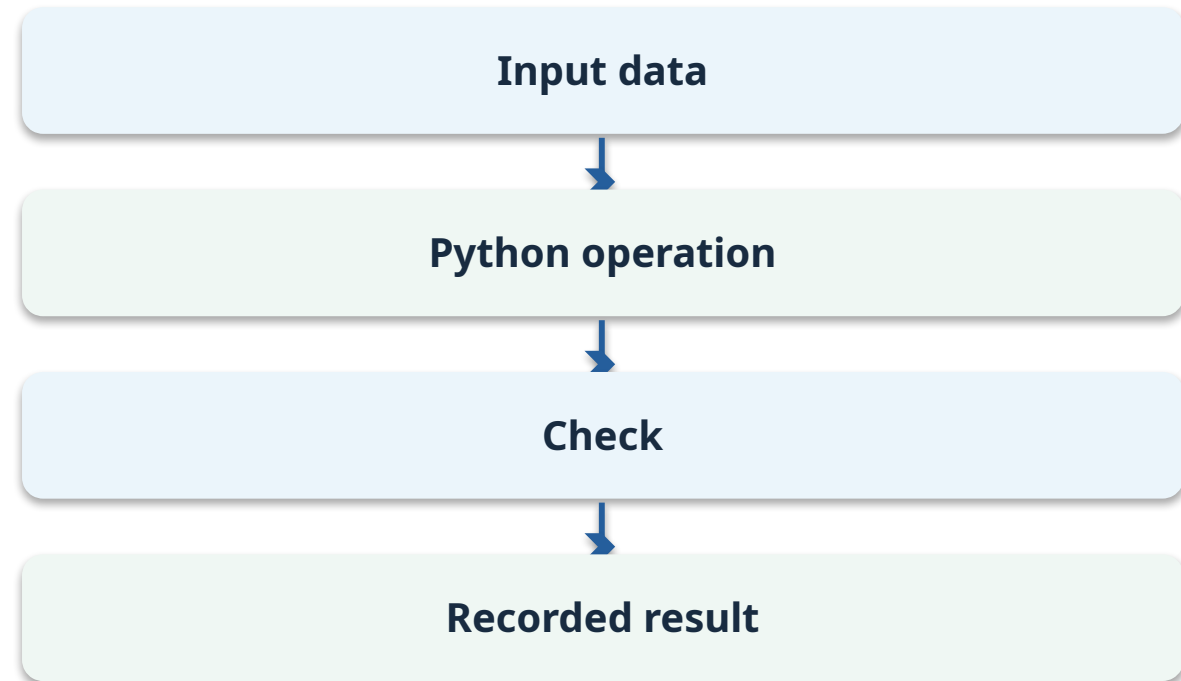
Compare top-1 and top-2 on practice cases

Find the input and check in Python

PRACTICE / WEEK 07

Locate the supplied input, the transformation, the checking line and the printed result.

FOLLOW THE DATA THROUGH EACH ARROW



Predict one controlled change

PRACTICE / WEEK 07

Replace a relevant passage with an irrelevant one and observe abstention.

What should change in the result? What stays fixed?

Before: original case

After: one change

Keep all other inputs fixed.

Run, compare and record

PRACTICE / WEEK 07

Record the input, the one changed setting, the observed result and a reason for any difference.

Input	Setting	Result
Original	baseline	record
Changed	one edit	compare

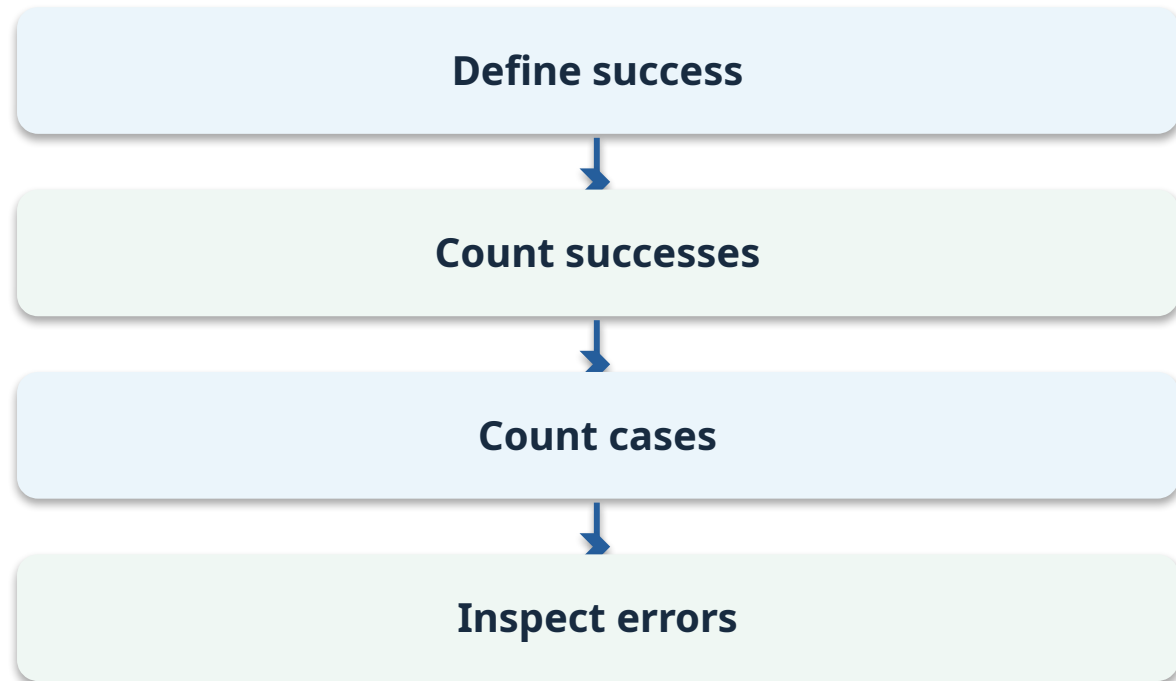
Measure without hiding the denominator

CHECK / WEEK 07

Supported claims / checked claims and no-answer accuracy

Write the numerator, denominator and one failure case.

FOLLOW THE DATA THROUGH EACH ARROW



Error detective: find the layer

CHECK / WEEK 07

Plausible wrong claim: A citation label proves that the cited passage supports the answer.

Correct it using the case evidence.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

A citation label proves that the cited passage supports the answer.

SUPPORTED EXPLANATION

Read the passage and check that it entails the claim; a plausible-looking source tag can be wrong.

Name a test that would catch this failure.

Transfer to a new situation

CHECK / WEEK 07

A fresh question asks about a rule absent from all five policies. What should the assistant do?

Explain which source or test would settle it.

VOTE FIRST • GIVE A REASON

A Use the visible evidence

B Flag missing or unsafe evidence

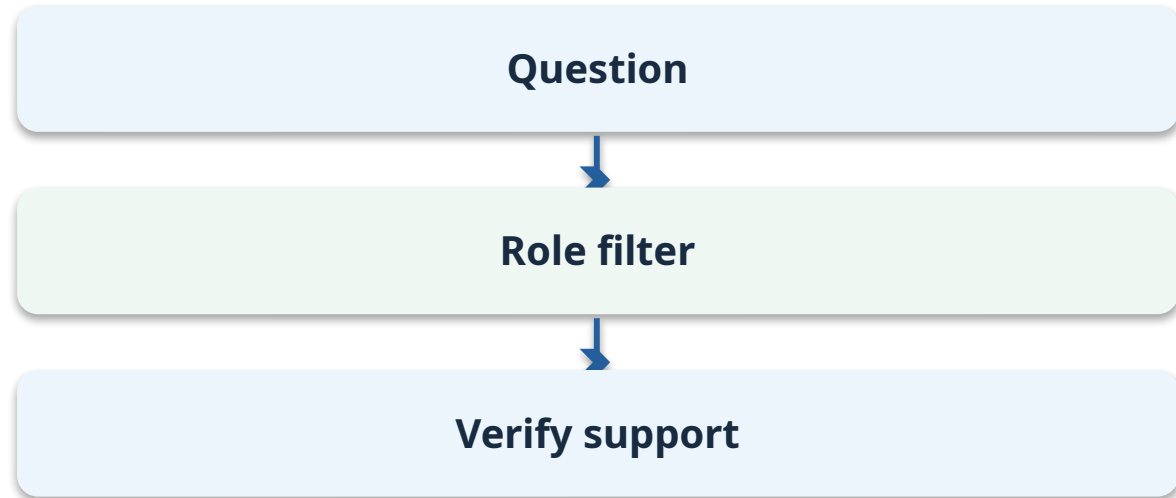
Which source or test could settle this?

Explain it to a partner

CHECK / WEEK 07

Without reading the slide, name the input, one processing step and the check. Your partner should ask “How do you know?”

FOLLOW THE DATA THROUGH EACH ARROW



Exam rehearsal · explain and apply

CHECK / WEEK 07

Why reserve final questions?

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

Compare with the Word guide

Exam rehearsal · explain and apply

CHECK / WEEK 07

Two of four material claims are supported. What is citation support?

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

Compare with the Word guide

What to hand in

PRACTICE / WEEK 07

Twenty-case practice table; ten final questions remain sealed for assessment.

Include the one changed input and an error explanation.

1

Original run

2

One changed input

3

Comparison

4

Explanation + source

Three ideas to recall next week

CHECK / WEEK 07

Retrieve, answer, check each claim, and abstain without evidence.

Read the linked official sources in the Word guide and redraw one diagram from memory.

1

RAG

2

Citation support

3

No-answer case

4

Test split